

Unit – 4 : Session – 7 : SLO – 2 | Execute Parallel Tasks with Process Pools

I. Programming Task

1. `from multiprocessing import Pool`
3. `def cube(x):`
4. `return x ** 3`
6. `if __name__ == "__main__":`
7. `with Pool(3) as p:`
8. `print(p.map(cube, [1, 2, 3])) # [1, 8, 27]`

II. Explanation

1. `Pool(3)` creates three worker processes; `map()` sends each number to a worker and collects the cubes.
2. The `__main__` guard is required so child processes (spawned/forked) do not re-run the pool creation code.
3. Output: `[1, 8, 27]` — computed in parallel across the worker processes.